JAVA

Model	**NDCG@10**
Bm25	0.4438
TFIDF	0.4255
Dirichlet	0.3250

Python

Model	**NDCG@10**
BM25	0.4630
TFIDF	0.4505
Dirichlet	0.3392

Javascript

Model	**NDCG@10**
BM25	0.4732
TFIDF	0.4609
Dirichlet	0.3203

CODE

To make batches run the **make_bulk_batches.py** script by running:

python make_bulk_batches.py <lang> <target_index>
EXAMPLE - python make_bulk_batches.py python_tfidf

<target_index> -> ("python_bm25", "BM25"),

("python_tfidf", "TF-IDF"),

("python_dirichlet", "Dirichlet"),

Curl:

Using the curl command appropriately

Example:

curl.exe -s -H "Content-Type: application/json" -XPOST "http://localhost:9200/_bulk" --data-binary "@Assignment2/cs589assignment1/dataset/python/python_bulk_0001.json"

Python:

BM25

1. PUT **python_bm25**
2. {
3. "settings": {
4. "analysis": {
5. "analyzer": {
6. "my_analyzer": { "tokenizer": "whitespace", "filter": ["lowercase","porter_stem"] }
7. }
8. },
9. "similarity": {
10. "default": { "type": "BM25", "k1": 2.0, "b": 0.6 }
11. }
12. },
13. "mappings": { "properties": { "title": { "type": "text", "analyzer": "my_analyzer" } } }
14. }

```
PUT python_dirichlet
{
 "settings": {
  "analysis": {
   "analyzer": {
    "my_analyzer": { "tokenizer": "whitespace", "filter": ["lowercase","porter_stem"] }
   }
  },
  "similarity": { "default": { "type": "LMDirichlet" } }
 },
 "mappings": { "properties": { "title": { "type": "text", "analyzer": "my_analyzer" } } }
}
```

```
PUT python_tfidf
{
 "settings": {
  "analysis": {
   "analyzer": {
    "my_analyzer": { "tokenizer": "whitespace", "filter": ["lowercase","porter_stem"] }
   }
  },
  "similarity": {
```

```
    "default": {
     "type": "scripted",
     "script": {
      "source": "\n          double tf  = Math.sqrt(doc.freq);\n          double idf = Math.log((field.docCount + 1.0) / (term.docFreq + 1.0)) + 1.0;\n          double norm = 1.0 / Math.sqrt(doc.length);\n          return query.boost * tf * idf * norm;\n        "
     }
    }
   }
  },
  "mappings": { "properties": { "title": { "type":  "text", "analyzer": "my_analyzer" } } }
}
```

JavaScript

```
PUT js_bm25
{
"settings": {
"analysis": {
"analyzer": {
"my_analyzer": { "tokenizer": "whitespace", "filter": ["lowercase","porter_stem"] }
}
},
"similarity": { "default": { "type": "BM25", "k1": 2.0, "b": 0.6 } }
},
"mappings": { "properties": { "title": { "type": "text", "analyzer": "my_analyzer" } } }
}
```

```
PUT js_tfidf
{
  "settings": {
   "analysis": {
    "analyzer": {
     "my_analyzer": { "tokenizer": "whitespace", "filter": ["lowercase","porter_stem"] }
    }
   },
   "similarity": {
    "default": {
     "type": "scripted",
     "script": {
      "source": "\n          double tf  = Math.sqrt(doc.freq);\n          double idf = Math.log((field.docCount + 1.0) / (term.docFreq + 1.0)) + 1.0;\n          double norm = 1.0 / Math.sqrt(doc.length);\n          return query.boost * tf * idf * norm;\n        "
     }
    }
   }
  },
  "mappings": { "properties": { "title": { "type": "text", "analyzer": "my_analyzer" } } }
}
```

```
PUT js_dirichlet
{
```

```
  "settings": {
    "analysis": {
      "analyzer": {
        "my_analyzer": { "tokenizer": "whitespace", "filter": ["lowercase","porter_stem"] }
      }
    },
    "similarity": { "default": { "type": "LMDirichlet" } }
  },
  "mappings": { "properties": { "title": { "type": "text", "analyzer": "my_analyzer" } } }
}
```

JAVA:

BM25

1. PUT java_bm25
2. {
3. "settings": {
4. "analysis": {
5. "analyzer": {
6. "my_analyzer": { "tokenizer": "whitespace", "filter": ["lowercase","porter_stem"] }
7. }
8. },
9. "similarity": { "default": { "type": "BM25", "k1": 2.0, "b": 0.6 } }
10. },
11. "mappings": { "properties": { "title": { "type": "text", "analyzer": "my_analyzer" } } }
12. }

Dirichlet

1. PUT java_dirichlet
2. {
3. "settings": {
4. "analysis": {
5. "analyzer": {
6. "my_analyzer": { "tokenizer": "whitespace", "filter": ["lowercase","porter_stem"] }
7. }
8. },
9. "similarity": { "default": { "type": "LMDirichlet" } }
10. },
11. "mappings": { "properties": { "title": { "type": "text", "analyzer": "my_analyzer" } } }
12. }

TFIDF

1. PUT java_tfidf
2. {
3. "settings": {
4. "analysis": {
5. "analyzer": {
6. "my_analyzer": { "tokenizer": "whitespace", "filter": ["lowercase","porter_stem"] }
7. }
8. },
9. "similarity": {

10. "default": {
11. "type": "scripted",
12. "script": {
13. "source": "\n   double tf = Math.sqrt(doc.freq);\n   double idf = Math.log((field.docCount + 1.0) / (term.docFreq + 1.0)) + 1.0;\n   double norm = 1.0 / Math.sqrt(doc.length);\n   return query.boost * tf * idf * norm;\n   "
14. }
15. }
16. }
17. },
18. "mappings": { "properties": { "title": { "type": "text", "analyzer": "my_analyzer" } } }
19. }